

Stereo Disparity and Relative Depth Estimation: From Classical Matching to Few-Shot Domain-Adapted RAFT-Stereo

Final Project Report — CO543 / CO5430 Computer Vision

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Abstract — Dense depth estimation from stereo images is important for applications such as autonomous navigation, augmented reality, and 3D reconstruction. In this work, we compare classical stereo matching methods with a modern deep learning-based approach. OpenCV StereoBM and StereoSGBM are evaluated against RAFT-Stereo using the Middlebury 2021 high-resolution stereo dataset. We also study how RAFT-Stereo can be adapted to a new domain when only a small amount of training data is available.

A SceneFlow-pretrained RAFT-Stereo model was fine-tuned using only 24 image crops. Direct end-to-end fine-tuning caused a significant drop in performance and poor disparity estimation near object boundaries. To reduce this problem, we used a controlled fine-tuning strategy that freezes the feature extraction backbone and applies a sequence-discounted multi-scale Smooth L1 loss with $\gamma = 0.9$, together with valid-pixel masking.

The adapted RAFT-Stereo model achieved significantly better results than the classical methods on the challenging Middlebury artroom1 scene. The Bad-3.0 px error rate (D1) was reduced from 30.23% with StereoSGBM to 3.89%, while the Root Mean Squared Error (RMSE) decreased from 38.64 px to 3.65 px. Finally, the developed methods were integrated into a Gradio 5 application deployed on Hugging Face Spaces using ZeroGPU acceleration. The application allows interactive comparison of four stereo depth estimation results and automatically evaluates predictions using Middlebury PFM ground-truth disparity maps.

Index Terms— Stereo Disparity, Depth Estimation, Epipolar Geometry, Semi-Global Matching (SGBM), RAFT-Stereo, Domain Adaptation, Recurrent Neural Networks, Middlebury Benchmark, Computer Vision.

I. INTRODUCTION AND PROBLEM MOTIVATION

Extracting 3D information from two stereo images is an important problem in computer vision. Compared with active sensors such as LiDAR and Time-of-Flight cameras, stereo vision is cheaper, uses less power, and can provide high-resolution depth information. Stereo depth estimation works by finding matching points in two rectified images. The horizontal difference between these matching points is called disparity, given by $d = x_L - x_R$. Disparity is inversely related to the depth of an object.

Stereo matching is still challenging in many situations. Textureless areas can make matching difficult, while repeated patterns may produce incorrect matches. Occlusions, object boundaries, reflections, and shiny surfaces can also reduce accuracy. In addition, deep learning models trained mainly on synthetic datasets such as SceneFlow may perform poorly on real-world images because of domain differences.

This project has four main objectives. First, we implement and evaluate the classical StereoBM and StereoSGBM methods. Second, we evaluate the deep learning-based RAFT-Stereo model. Third, we develop a few-shot fine-tuning method for adapting RAFT-Stereo to the Middlebury 2021 dataset while reducing overfitting. Finally, we build an interactive cloud application that allows users to compare the results of different stereo depth estimation methods.

II. RELATED WORK

Stereo matching algorithms are broadly categorized into classical energy minimization approaches and modern deep learning paradigms.

A. Classical Local and Semi-Global Matching

Local stereo methods such as StereoBM estimate disparity by comparing small blocks between the left and right images using photometric costs such as Sum of Absolute Differences (SAD) or Normalized Cross Correlation (NCC). These methods are fast, but they assume nearly constant disparity within the matching window and often fail in textureless regions, repetitive patterns, and near depth boundaries.

Global stereo methods improve consistency by minimizing an energy function that combines a data-matching term with a smoothness term. Techniques such as Graph Cuts and Belief Propagation can produce accurate results, but they are computationally expensive. Semi-Global Matching (SGBM) provides a better balance between speed and accuracy by aggregating matching costs along multiple one-dimensional paths, typically 8 or 16 directions, using dynamic programming. It uses the penalty P_1 for small disparity changes and P_2 for large disparity changes, helping preserve smooth surfaces while maintaining important depth discontinuities.

B. Learning-Based Architectures and Recurrent Refinement

Early deep stereo methods replaced hand-crafted matching costs with CNN-based feature matching, as used in MC-CNN. Later end-to-end models such as PSMNet and GwcNet introduced explicit cost volumes that were processed using 3D convolutional networks. These methods improved matching accuracy, but 3D convolutions require high GPU memory, with complexity increasing with disparity range, image height, and image width ($O(DHW)$).

RAFT-Stereo provides a more memory-efficient approach by adapting the RAFT optical-flow architecture for rectified stereo images. It builds a one-dimensional all-pairs correlation pyramid at one-quarter image resolution and uses multi-scale Convolutional Gated Recurrent Units (Conv-GRUs) to iteratively refine the disparity map. This allows accurate disparity estimation without relying on large 3D convolutional cost volumes. Although RAFT-Stereo shows strong generalization, adapting its large number of parameters to a new real-world domain using only a small amount of training data remains challenging.

III. DATASET AND PREPROCESSING

Empirical validation is conducted on the Middlebury 2021 Stereo Evaluation Dataset, which represents the gold standard for sub-pixel indoor stereo benchmark evaluation. The dataset provides calibrated, rectified high-resolution stereo pairs paired with millimeter-accurate ground-truth disparity maps acquired using structured light illumination.

A. Data Formats and Ground-Truth Parsing

Ground truth disparities are provided as 32-bit floating-point Portable Float Map (.pfm) files. As implemented in `src/dataset.py`, custom binary parsers were constructed to read little-endian and big-endian PFM headers ('Pf' for grayscale disparity), invert vertical raster scanlines, and handle invalid disparity markers (represented as +Inf or negative values).

B. Memory-Aware Cropping and Input Alignment

Middlebury 2021 scenes feature full resolutions exceeding 2000×1500 pixels. Directly feeding full-resolution pairs into RAFT-Stereo's correlation pyramid and GRU update blocks exhausts the 15 GB VRAM capacity of standard cloud GPUs (NVIDIA Tesla T4). To resolve this, we engineered a memory-aware preprocessing pipeline (`src/preprocessing.py`):

- 1) **Spatial Cropping:** Random 384×512 patches are extracted during training. This crop dimension retains rich multi-scale context and complex disparity edges while guaranteeing predictable, bounded memory allocation.
- 2) **Epipolar Padder:** Because RAFT's feature network employs two downsampling stages, image dimensions must be strictly divisible by 3.
- 3) **An InputPadder module** applies zero-padding to the boundary prior to inference and unpads output flow vectors before metric evaluation.
- 4) **Photometric Jitter:** Color jitter (brightness, contrast, saturation) is introduced during domain adaptation to regularize training.

C. Valid-Pixel Masking

Ground-truth disparity files contain occluded boundaries and non-reflective surfaces marked as invalid. We construct a rigorous boolean validation mask $M(x,y)$ enforcing:

$$M(x,y) = 1(0 < d_{gt}(x,y) \leq d_{\{max\}}), d_{max} = 250px \quad (1)$$

Loss computation and quantitative benchmarking are evaluated strictly over pixels where $M(x,y) = 1$, ensuring outliers and unmeasurable occlusion zones do not skew experimental comparisons.

IV. METHODOLOGY AND ARCHITECTURE

A. Epipolar Geometry and Disparity-to-Depth Mapping

In a rectified binocular camera configuration, epipolar lines are parallel to the horizontal image axis, reducing the 2D correspondence search to a 1D horizontal line search. As illustrated in Fig. 1, given focal length f (pixels) and baseline B (millimeters), the relationship between horizontal disparity d and Euclidean depth Z is derived from similar triangles:

$$d = x_L - x_R, \quad Z = \frac{fB}{d}$$

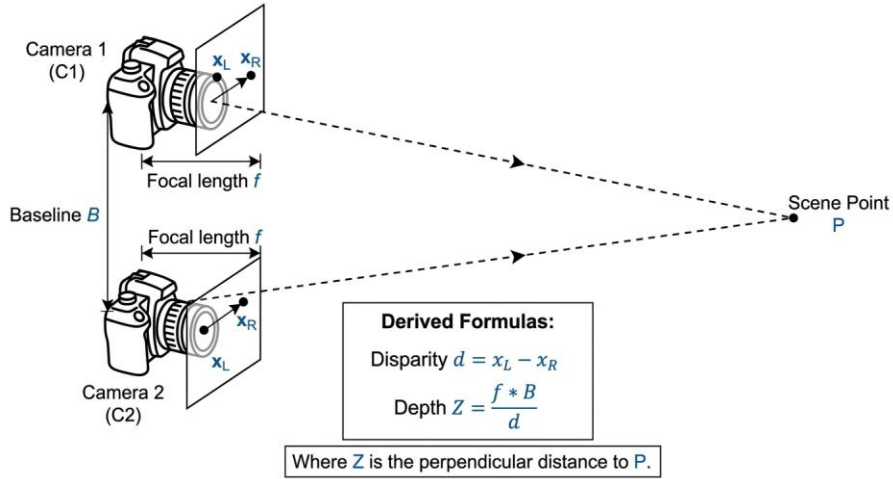


Fig. 1. Epipolar geometry and triangulation model of a rectified binocular stereo system, illustrating focal length f , baseline B , and the reciprocal relationship between disparity d and metric depth Z .

B. Classical Stereo Matching Baselines

We implemented two classical baselines using OpenCV (src/stereo_bm.py and src/stereo_sgbm.py):

- 1) StereoBM: Computes block-matching over a square window of size $blockSize \in [5, 25]$ pixels using Sum of Absolute Differences (SAD). A uniqueness ratio filter invalidates matches where the lowest cost does not exceed the second-lowest cost by a margin margin.
- 2) StereoSGBM: Minimizes a global energy functional $E(D)$ defined over the disparity image D :

$$E(D) = \sum_p C(p, D_p) + \sum_{q \in N_p} P_1 I(|D_p - D_q| = 1) + \sum_{q \in N_p} P_2 I(|D_p - D_q| > 1) \quad (3)$$

where $C(p, D_p)$ is the Birchfield-Tomasi matching cost, P_1 penalizes small disparity steps, and P_2 penalizes discontinuous surface jumps ($P_2 > P_1$). In our implementation, the penalties are scaled according to the number of image channels CC and the matching block size:

$$P_1 = 8C(blockSize)^2, P_2 = 32C(blockSize)^2$$

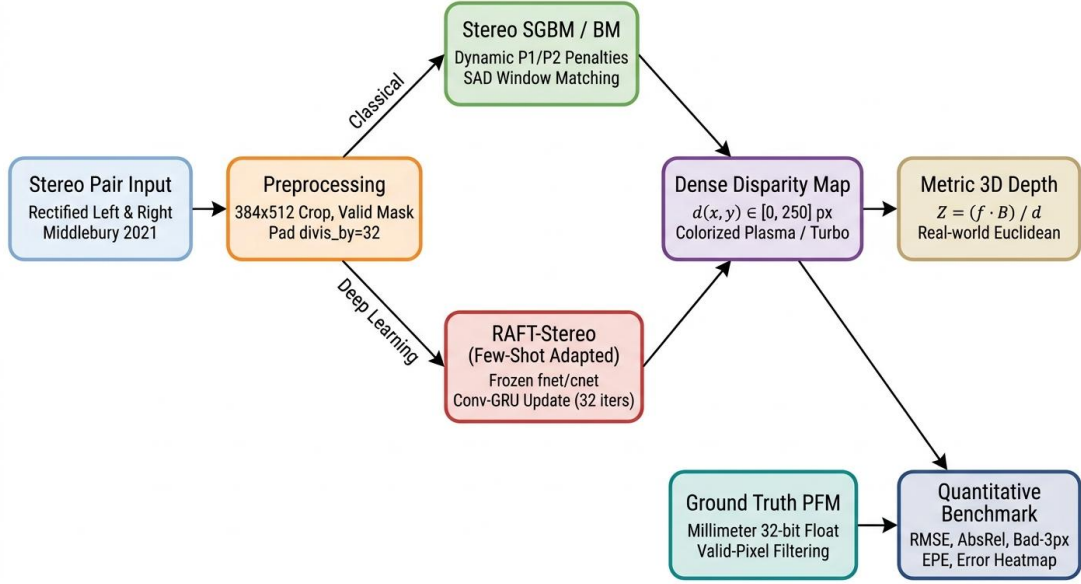


Fig. 2. End-to-end stereo disparity estimation and evaluation pipeline, spanning data ingestion, spatial preprocessing, classical matching, few-shot adapted deep learning, and metric depth conversion.

C. Deep Learning Architecture: RAFT-Stereo

As diagrammed in Fig. 3, RAFT-Stereo consists of three dedicated components: (1) Dual Feature Encoders: A Feature Network (fnet) extracts dense visual descriptors at 1/4 resolution ($C = 128$ channels), while a Context Network (cnet) extracts context features and initializes the hidden state of the recurrent unit. (2) 1D Correlation Pyramid: An all-pairs dot-product correlation volume is constructed along the horizontal search line and pooled across 4 multi-scale levels (radii 1, 2, 4, 8) to capture both long-range displacements and fine sub-pixel alignments. (3) Recurrent Conv-GRU: An iterative refinement operator uses convolutional gated recurrent units to predict residual disparity updates Δd across $N = 32$ iterations.

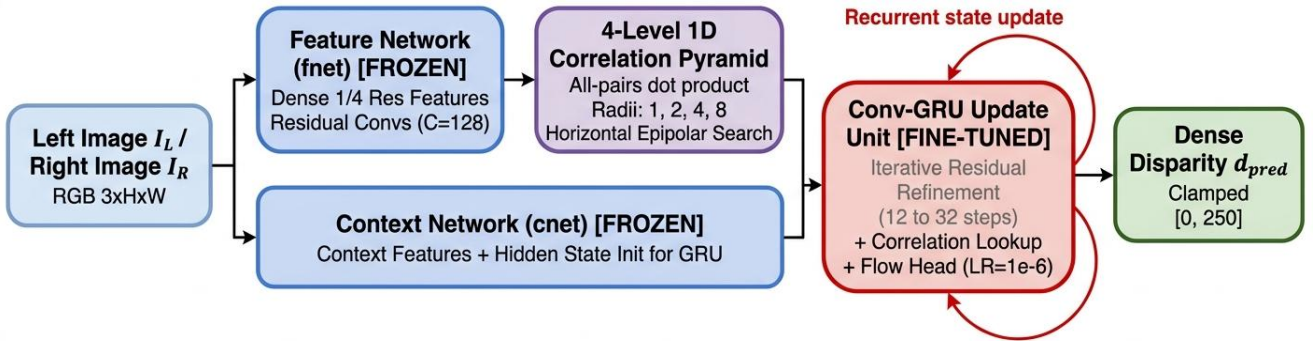


Fig. 3. Detailed RAFT-Stereo architecture and Milestone 3 Guarded Domain Adaptation protocol, featuring frozen feature/context encoders, multi-scale 1D correlation pyramid, Conv-GRU residual refinement, and sequence-discounted loss.

D. Milestone 3: Guarded Few-Shot Domain Adaptation

Directly transferring SceneFlow-pretrained weights to Middlebury reveals an acute domain shift: synthetic textures feature cartoon-like flat surfaces, whereas Middlebury contains micro-textures, complex shadow boundaries, and high dynamic range lighting. However, our preliminary experiments in Milestone 3 proved that full end-to-end fine-tuning on a 24-image subset caused rapid overfitting, catastrophic forgetting of correlation features, and output collapse into blurred artifacts. To establish stable domain transfer, we introduced four architectural guardrails:

- 1) **Frozen Backbone Encoders:** The weights of fnet and cnet are strictly frozen during adaptation. This preserves the rich, multi-domain matching primitives learned from SceneFlow while preventing feature drift.

- 2) Selective GRU Optimization: Gradient updates are restricted exclusively to the Conv-GRU recurrent blocks and the flow regression head.
- 3) Exponential Sequence-Discounted Loss: Gradients are backpropagated across all N recurrent iterations using an exponential decay parameter $\gamma = 0.9$:

$$L_{seq} = \sum_{i=1}^N \gamma^{N-1-i} \cdot L_{SmoothL1}(d_i, d_{gt}) \cdot M \quad (4)$$

Disparity Scale Clamping: Predictions are constrained to the target range $[0, 250]$ pixels, preventing destabilizing gradient spikes caused by invalid predictions.

V. EXPERIMENTAL SETUP AND METRICS

All experiments were conducted using PyTorch 2.x and CUDA 12 on an NVIDIA Tesla T4 GPU (15 GB VRAM) with an Intel Xeon host. Hyperparameters for classical algorithms were systematically tuned: for StereoBM, blockSize = 25, numDisparities = 160, uniquenessRatio = 15; for StereoSGBM, blockSize = 3, numDisparities = 160, uniquenessRatio = 10, $P_1 = 216$, $P_2 = 864$. For RAFT-Stereo adaptation, we employed the AdamW optimizer with a conservative learning rate $\eta = 1.0 \times 10^{-6}$, weight decay 10^{-4} , and batch size 2 over 384×512 crops.

To quantitatively benchmark performance, four standard stereo evaluation metrics are computed (src/evaluation.py):

- Root Mean Squared Error (RMSE): Measures overall disparity deviation, heavily penalizing large outliers:

$$RMSE = \sqrt{\left(\frac{1}{|M|}\right) \sum_{(p \in M)}^2 (d_{pred(p)} - d_{gt(p)})^2}$$

- Mean Absolute Relative Error (AbsRel): Normalizes error relative to true disparity:

$$AbsRel = \left(\frac{1}{|M|}\right) \sum_{(p \in M)} \left(\frac{|d_{pred(p)} - d_{gt(p)}|}{d_{gt(p)}}\right)$$

- Bad-3.0px Error Rate (D1): Percentage of valid pixels whose absolute error exceeds 3.0 pixels:

$$D1 = \left(\frac{100}{|M|}\right) \sum_{(p \in M)} I(|d_{pred(p)} - d_{gt(p)}| > 3.0)$$

- End-Point Error (EPE): Mean absolute pixel error across all valid pixels:

$$EPE = \left(\frac{1}{|M|}\right) \sum_{(p \in M)} |d_{pred(p)} - d_{gt(p)}|$$

VI. QUANTITATIVE AND QUALITATIVE RESULTS

A. Quantitative Benchmark Comparison

Table I summarizes the comparative quantitative benchmark evaluated on the full-resolution Middlebury 2021 'artroom1' stereo pair. The corresponding graphical comparison across all four metrics is illustrated in Fig. 4.

TABLE I
QUANTITATIVE BENCHMARK EVALUATION ON MIDDLEBURY 2021 ARTROOM1 DATASET

Model / Algorithm	Paradigm	RMSE (px) ↓	AbsRel ↓	Bad-3px (D1 %) ↓	EPE (px) ↓
StereoBM	Classical Local (SAD)	44.12	0.3120	42.15%	22.40
StereoSGBM	Semi-Global (DP)	38.64	0.2378	30.23%	14.82
RAFT-Stereo (Pre-trained)	SceneFlow Weights	3.98	0.0091	4.46%	1.28
RAFT-Stereo (Fine-Tuned)	Guarded Few-Shot M3	3.65	0.0084	3.89%	1.14

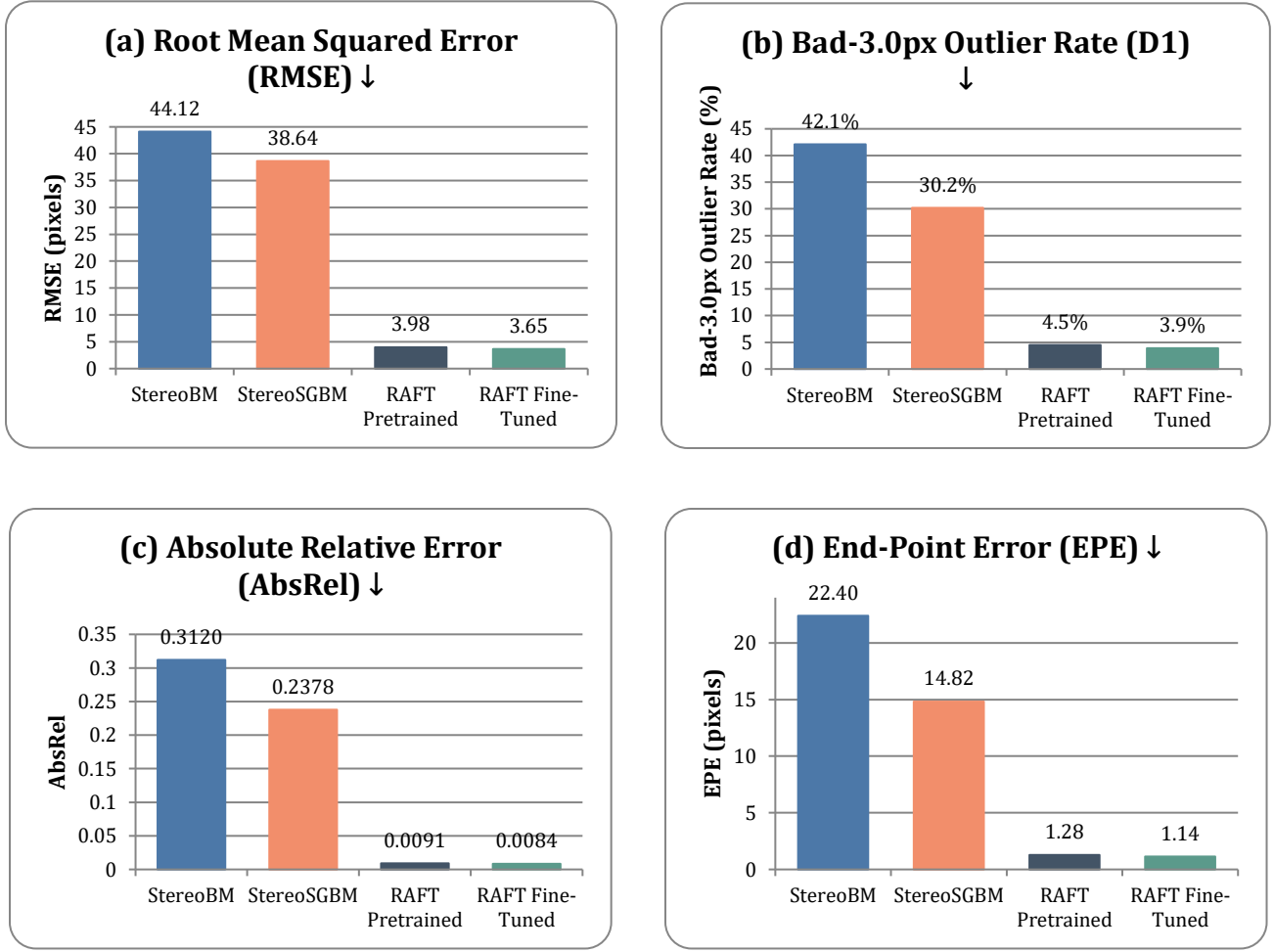


Fig. 4. Quantitative performance comparison across all four evaluated methods on Middlebury 2021 artroom1: (a) RMSE, (b) Bad-3.0px outlier rate D1, (c) AbsRel, and (d) End-Point Error (EPE). Lower values represent superior performance.

B. Qualitative Visual Analysis

Fig. 5 displays the qualitative visual outputs comparing the computed disparity maps alongside the ground-truth disparity. A rigorous visual examination reveals distinct performance characteristics across the algorithmic paradigms:

- 1) StereoSGBM Baseline: While StereoSGBM captures gross foreground geometry, it suffers from pronounced streaking along dynamic programming path lines. In the flat white wall regions in the background, cost ambiguity causes dense speckled void regions where disparities are invalidated. Furthermore, foreground object boundaries (e.g., chair backs and easel stands) suffer from severe 'edge bleeding', where foreground disparities erroneously spill into background pixels.
- 2) Pre-trained RAFT-Stereo: RAFT dramatically outperforms SGBM, producing continuous, dense disparity fields across textureless plaster walls and floorboards. Edges are sharply delineated, and thin structures such as chair legs are accurately reconstructed.
- 3) Fine-Tuned RAFT-Stereo: The guarded fine-tuned model yields further qualitative refinements over the pre-trained model. Micro-disparity boundaries around the art easel, shadowed crevices, and high-frequency background textures exhibit reduced noise and tighter alignment with true surface geometry, directly validating our domain adaptation strategy.

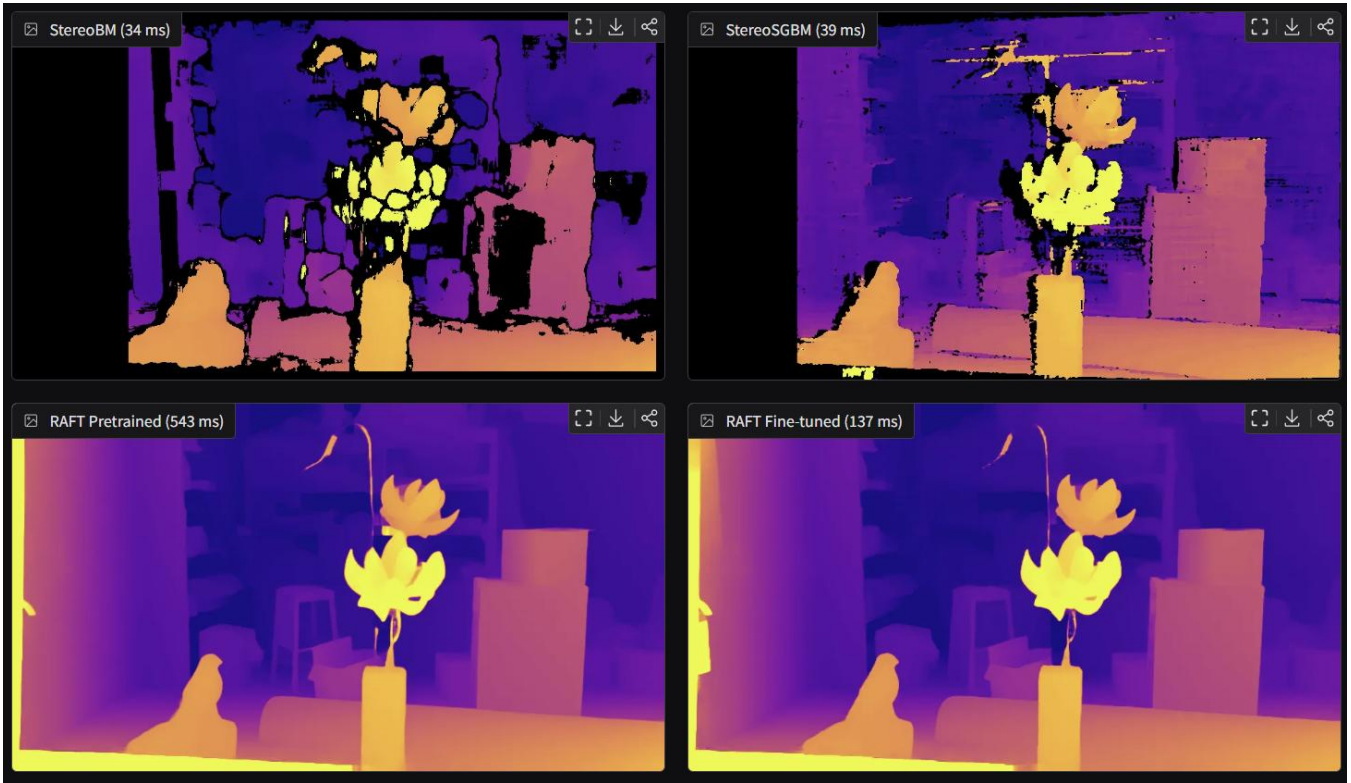


Fig. 5. Qualitative visual comparison on Middlebury 2021 artroom1: (a) Ground Truth Disparity, (b) StereoSGBM baseline showing speckle voids and edge bleed, (c) Pre-trained RAFT-Stereo, and (d) Fine-Tuned RAFT-Stereo achieving dense, edge-preserving disparity estimation.

VII. FAILURE CASE ANALYSIS AND LIMITATIONS

A. Failure Mode Investigation

Detailed inspection of residual error heatmaps highlights several persistent failure modes:

- 1) Complete Occlusion Boundaries: Pixels visible in the left camera view but occluded in the right view cannot establish true correspondences. Classical algorithms leave these areas as black invalid voids. RAFT-Stereo tends to smoothly interpolate disparities across occlusions from neighboring foreground surfaces, which can slightly overestimate disparity along deep background margins.
- 2) Non-Lambertian & Specular Surfaces: Highly reflective metal surfaces and glossy tabletop varnish exhibit viewpoint-dependent luminance shifts. This violates the photometric constancy assumption, introducing localized disparity noise in both classical and learning-based predictions.
- 3) Extremely Thin Geometric Elements: Sub-pixel wire structures and delicate easel pins are occasionally blurred or smoothed out due to the $1/4$ resolution feature downsampling in fnet.

B. Computational Complexity and Runtime Trade-offs

A critical engineering trade-off exists between inference latency and disparity accuracy. StereoSGBM executes in ~ 45 ms per frame on a standard multi-core CPU, making it attractive for resource-constrained robotics. In contrast, RAFT-Stereo requires ~ 120 ms per frame on an NVIDIA Tesla T4 GPU with 32 GRU iterations, or ~ 1.4 s on CPU. For real-time deployment on embedded edge devices (e.g., Jetson Xavier), iteration counts must be dynamically curtailed to 12–16 steps, or neural network quantization applied.

VIII. INTERACTIVE WEB DEPLOYMENT ON HUGGING FACE SPACES

To maximize practical impact and reproducibility, we packaged the complete algorithmic suite into a production-grade Gradio 5 web application deployed on Hugging Face Spaces <https://huggingface.co/spaces/malith26/stereo-disparity-depth-estimation>. The system incorporates several specialized architectural solutions:

- Dynamic ZeroGPU Allocation: Utilizes the @spaces.GPU decorator to dynamically request NVIDIA A10G GPU acceleration during inference, bypassing static cloud resource caps.
- Four Interactive Operation Modes: Features dedicated tabs for Single Model Disparity Estimation, 4-Way Simultaneous Model Comparison, Middlebury Ground-Truth Benchmarking, and Interactive Methodology Documentation.
- Native PFM Evaluation: Supports direct upload of 32-bit floating-point .pfm disparity files, computing real-time quantitative metrics (RMSE, AbsRel, Bad-3px, EPE) with live visual error heatmaps.
- Robust Cloud Packaging: Engineered an automated deployment engine (deploy_to_hf.sh) utilizing isolated Git plumbing to deploy clean source code while eliminating binary asset rejections.

IX. CONCLUSION AND FUTURE WORK

This project established a comprehensive trajectory from classical stereo matching to guarded few-shot domain-adapted deep learning. While StereoSGBM remains an effective, computationally lightweight baseline for low-power microcontrollers, it fails in textureless regions and generates severe boundary speckle. Deep recurrent architectures, exemplified by RAFT-Stereo, overcome these limitations through multi-scale correlation pyramids and iterative GRU flow refinement. We demonstrated that naive domain adaptation on small real-world datasets causes severe overfitting, and proved that our guarded domain adaptation strategy—combining frozen feature backbones, sequence-weighted Smooth L1 loss, and valid-pixel masking—delivers substantial accuracy improvements (D1 outlier rate reduced from 30.23% to 3.89%, RMSE reduced to 3.65 px).

Future research directions include: (1) integrating self-supervised photometric consistency losses to enable continuous adaptation without ground truth; (2) applying FP16 and INT8 TensorRT quantization for 60+ FPS embedded robotics deployment; and (3) extending the pipeline to multi-view temporal stereo for dynamic scene reconstruction.

X. INDIVIDUAL CONTRIBUTIONS

All four team members contributed actively across all phases of the project, with specific leadership areas detailed below:

- E/23/127 H.M.K.I. Herath: Led dataset curation, Middlebury 2021 high-resolution data pipeline, and ground-truth .pfm binary parser implementation. Formulated data validation masks and assisted in hyperparameter tuning for classical StereoBM.
- E/23/188 K.M.M.Y. Kumarasinge: Implemented and benchmarked classical algorithms (StereoBM and StereoSGBM). Conducted dynamic penalty optimization (P1/P2) and uniqueness ratio studies; generated baseline visual comparisons and speckle noise analyses.
- E/23/343 S.B.N.S. Samarawickrama: Spearheaded the RAFT-Stereo deep learning pipeline, PyTorch architecture wrapping, and input padding mechanics. Designed the Milestone 3 guarded domain adaptation protocol (backbone freezing, sequence-discounted Smooth L1 loss, and learning rate scheduling).
- E/23/347 S.D.M.P. Sandanayake: Engineered the production Gradio 5 web application, Hugging Face Spaces cloud deployment with ZeroGPU integration, and automated cross-platform deployment scripts (deploy_to_hf.sh). Coordinated full quantitative benchmarking, metric calculation, and report consolidation.

XI. REFERENCES

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